# Romy Mondello

**AI AND FULL-STACK ENGINEER - AGENT INFRASTRUCTURE - DEVELOPER TOOLS**

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AI, full-stack, and systems engineer with six years of professional software-engineering experience since joining Knox Financial in 2020. I build agent systems and the infrastructure they run on, with public work spanning JEV workflows for Codex and Claude Code, QQ runtime contributions, Rust systems, and formal verification. Earlier independent AI work began with deep learning in 2017 and language models around 2019, including GPT-3 beta participation.

## Independent product engineering and open source

### Independent Product Engineering / Open Source — May 2025 - present

- Built **JEV Workflows**, a public open-source plugin for Codex and Claude Code that provides typed decision consultation, failure diagnosis, and evidence-based completion review. Model recommendations remain advisory and separate from permission to act.
- Contributed to the open-source Rust QQ coding-agent harness in merged PRs #333, #319, and #72, covering session-history recovery after held-tool cancellation or restart, shared JEV protocol validation, review checkpoints, and provider-neutral reasoning-effort settings.
- Co-developing a bounded local agent-execution prototype with a cofounder. It explores Temporal coordination, PostgreSQL run state, isolated workers, explicit cancellation controls, persisted evidence, and guarded verification.
- Built Ferrite, a public Rust-first TypeScript application-framework prototype with server-side rendering, file-system routing, artifact serving, WASM and Node bindings, and browser hydration.

### Formal research — Independent

- Authored a 10-page arXiv preprint giving a dimension-two counterexample to the separable Jacobian conjecture in characteristic two, with Jacobian determinant 1 and separable degree 3. Formalized the full theorem in Lean 4 and Mathlib with digest-locked sources and replay receipts.

## Professional experience

### Further - Consultant, Research Engineer (contract) — Feb 10 - Apr 14, 2025

- Co-designed the LangGraph agent and data-pipeline architecture with the CTO, then built Zillow and mortgage API pipelines with typed state, tool orchestration, retrieval, and rule-based risk checks.
- Helped the team use Cursor and coding agents, and fixed LaTeX rendering and API timeouts in the AI Genius web app.

### Pillzumi - Machine Learning Engineer (contract) — Dec 2024 - Feb 2025

- Built long-running elizaOS social-agent integrations across WebRTC and X direct-message workflows, with persistent memory, action frameworks, and API-rate-limit handling.

### Verizon Business - Customer Solutions — Apr - Dec 2024

- Owned a CRM book of business in a technical-consultative B2B motion covering discovery, solution design, customer engineering conversations, and closing.

### Verus AI - Data Scientist — 2022 - 2023

- Built Google Cloud pipelines into BigQuery and Vertex AI, trained recurrent models for spend, usage, and customer-growth forecasting, and monitored model and data quality.

### Knox Financial - Backend Software Engineer — Jul 2020 - Mar 2021

- Built real-estate search on data feeds and AWS and Google APIs, distributed ingestion, PostgreSQL forecasting services, REST APIs, funnel instrumentation, and forecast visuals.
- Hardened CI/CD, reduced flaky-test noise, introduced deploy gates, and reviewed backend code.

**Education**

Associate of Science, Computer Science
Bunker Hill Community College - Dean's List

**Technical**

TypeScript, JavaScript, Python, Rust, Lean 4, Swift, Go | React, Next.js, Node.js, FastAPI, Axum | PostgreSQL, Temporal, Redis, Docker, AWS, GCP, BigQuery | LangGraph, OpenClaw, elizaOS, RAG, MLX, PyTorch/MPS